

SGA 2: Local Cohomology and Lefschetz Theorems

Exposé I, §2 — The functors $H_Z^*(X, F)$ and $\underline{H}_Z^*(F)$

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Exposé I

Global and Local Cohomological Invariants Relative to a Closed Subspace

2. The functors $H_Z^*(X, F)$ and $\underline{H}_Z^*(F)$

Definition 2.1. *The derived functors, in the variable F , of the functors $\Gamma_Z(F)$ and $\underline{\Gamma}_Z(F)$ are denoted by $H_Z^*(X, F)$ and $\underline{H}_Z^*(F)$, respectively.*

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They are cohomological functors with values, respectively, in the category of abelian groups and in the category of sheaves of abelian groups on X . When Z is closed, $H_Z^*(X, F)$ is, by definition, simply $H_\Phi^*(X, F)$, where Φ denotes the family of closed subsets of X contained in Z . When Z is open, we shall see that $H_Z^*(X, F)$ is simply $H^*(Z, F) = H^*(Z, F|_Z)$, by the following proposition.

Proposition 2.2 (Excision theorem). *Let V be an open subset of X containing Z . There is an isomorphism of cohomological functors in F :*

$$H_Z^*(X, F) \longrightarrow H_Z^*(V, F|_V). \quad (19)$$

Indeed, there is a functorial isomorphism $\Gamma_Z^X \simeq \Gamma_Z^V j^!$, where $j: V \rightarrow X$ is the inclusion and $j^!$ is therefore the restriction functor (cf. equation (14)). This last functor is exact and, by Corollary 1.4, takes injectives to injectives; the isomorphism (19) follows immediately.

When Z is open, we may take $V = Z$, and obtain:

Corollary 2.3. *Suppose that Z is open. There is an isomorphism of cohomological functors*

$$H_Z^*(X, F) = H^*(Z, F). \quad (20)$$

The isomorphisms of Corollary 1.6 and the definitions give the following (cf. [Tôhoku]).

Proposition 2.3 bis. *There are isomorphisms of cohomological functors*

$$H_Z^*(X, F) \simeq \text{Ext}^*(X; \mathbf{Z}_{Z,X}, F), \quad (21)$$

$$\underline{H}_Z^*(F) \simeq \underline{\text{Ext}}^*(\mathbf{Z}_{Z,X}, F). \quad (21 \text{ bis})$$

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Editorial note. This proposition is numbered 2.3 in the original edition; the corrected branch designates it 2.3 bis.

We may therefore apply the results of [Tôhoku] on the Ext of modules. First, we record the following interpretation of the sheaves $\underline{H}_Z^*(F)$ in terms of the global groups $H_Z^*(X, F)$.

Corollary 2.4. *$\underline{H}_Z^*(F)$ is canonically isomorphic to the sheaf associated with the presheaf*

$$U \mapsto H_{Z \cap U}^*(U, F|_U).$$

In particular, using Corollary 2.3, we obtain:

Corollary 2.5. *Suppose that Z is open. There is an isomorphism of cohomological functors*

$$\underline{H}_Z^*(F) = R^*i_*i^*(F), \quad (22)$$

where $i: Z \rightarrow X$ is the inclusion.

The Ext spectral sequence gives the following important spectral sequence.

Theorem 2.6. *There is a spectral sequence, functorial in F , abutting to $H_Z^*(X, F)$ and having initial term*

$$E_2^{p,q}(F) = H^p(X, \underline{H}_Z^q(F)). \quad (23)$$

Remarks 2.7. It follows immediately from Corollary 2.4 that the sheaves $\underline{H}_Z^q(F)$ vanish on $X - \overline{Z}$, and also vanish in the interior of Z when $q \neq 0$; hence, for such q , $\underline{H}_Z^q(F)$ is supported on the boundary of Z .

Consequently, the right-hand side of (23) may be interpreted as a cohomology group on $\overline{Z}$. We shall use Theorem 2.6 when Z is closed in X ; in that case the right-hand side of (23) may be interpreted as a cohomology group computed on Z :

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$$E_2^{p,q}(F) = H^p(Z, \underline{H}_Z^q(F)). \quad (23 \text{ bis})$$

The corresponding equation was numbered (23) in the original edition.

When Z is open, the spectral sequence of Theorem 2.6 is simply the Leray spectral sequence for the continuous map $i: Z \rightarrow X$, in view of the interpretation in Corollary 2.5 used to compute its initial term.

We return to the exact sequence (18). It gives rise to an exact Ext sequence (cf. [Tôhoku]). *Editorial note.* The original edition cited Proposition 1.10 at this point; the corrected branch cites equation (18).

Theorem 2.8. *Let Z be a locally closed subset of X , let Z' be a closed subset of Z , and put $Z'' = Z - Z'$. There is a long exact sequence, functorial in F :*

$$\begin{aligned} 0 \longrightarrow H_{Z'}^0(X, F) \longrightarrow H_Z^0(X, F) \longrightarrow H_{Z''}^0(X, F) \xrightarrow{\partial} H_{Z'}^1(X, F) \longrightarrow H_Z^1(X, F) \longrightarrow \cdots \\ \cdots \longrightarrow H_{Z'}^i(X, F) \longrightarrow H_Z^i(X, F) \longrightarrow H_{Z''}^i(X, F) \xrightarrow{\partial} H_{Z'}^{i+1}(X, F) \longrightarrow \cdots \end{aligned} \quad (24)$$

Recall how this exact sequence is obtained. Let $C(F)$ be an injective resolution of F . Then the exact sequence (18)¹ gives the exact sequence

$$0 \longrightarrow \Gamma_{Z'}(C(F)) \longrightarrow \Gamma_Z(C(F)) \longrightarrow \Gamma_{Z''}(C(F)) \longrightarrow 0. \quad (25)$$

This is precisely the sequence defined in Proposition 1.8. Its cohomology long exact sequence is (24).

Source note. The corrected French TeX prints $\Gamma(C(F))$ in the middle term of (25). Equation (24), the construction from (18), and Proposition 1.8 require $\Gamma_Z(C(F))$; the support subscript is restored here and the emendation is not silent.

The most important case for us is the one in which Z is closed (and we may always reduce to it by replacing X with an open subset V in which Z is closed). Then Z' is closed, Z'' is closed in the open subset $X - Z'$, and we may write

$$H_{Z''}^i(X, F) = H_{Z''}^i(X - Z', F|_{X-Z'}). \quad (26)$$

This permits us to write the exact sequence (24) in terms of cohomology with support in a given closed subset. The most common case is $Z = X$. Writing $Z' = A$ for simplicity, we obtain:

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Corollary 2.9. *Let A be a closed subset of X . There is a long exact sequence, functorial in F :*

$$\begin{aligned} 0 \longrightarrow H_A^0(X, F) \longrightarrow H^0(X, F) \longrightarrow H^0(X - A, F) \xrightarrow{\partial} H_A^1(X, F) \longrightarrow \cdots \\ \cdots \longrightarrow H_A^i(X, F) \longrightarrow H^i(X, F) \longrightarrow H^i(X - A, F) \xrightarrow{\partial} H_A^{i+1}(X, F) \longrightarrow \cdots \end{aligned} \quad (27)$$

This exact sequence shows that $H_A^i(X, F)$ plays the role of a relative cohomology group of X modulo $X - A$, with coefficients in F . It is in this capacity that it arises naturally in applications. By “sheafifying” (24) and (27), or by arguing directly, and taking into account that the sheaf associated with $U \mapsto H^i(U, F)$ is zero for $i > 0$, we obtain:

Corollary 2.10. *Under the hypotheses of Theorem 2.8 there is a long exact sequence, functorial in F :*

$$\cdots \longrightarrow \underline{H}_{Z'}^i(F) \longrightarrow \underline{H}_Z^i(F) \longrightarrow \underline{H}_{Z''}^i(F) \xrightarrow{\partial} \underline{H}_{Z'}^{i+1}(F) \longrightarrow \cdots \quad (24 \text{ bis})$$

Corollary 2.11. *Let A be a closed subset of X . There is an exact sequence, functorial in F :*

$$0 \longrightarrow \underline{H}_A^0(F) \longrightarrow F \longrightarrow f_*(F|_{X-A}) \xrightarrow{\partial} \underline{H}_A^1(F) \longrightarrow 0, \quad (28)$$

and, for $i \geq 2$, canonical isomorphisms

$$\underline{H}_A^i(F) = \underline{H}_{X-A}^{i-1}(F) = R^{i-1}f_*(F|_{X-A}), \quad (29)$$

¹See the preceding editorial note.

where $f: X - A \rightarrow X$ is the inclusion.

Thus $\underline{H}_A^0(F)$ and $\underline{H}_A^1(F)$ are defined, respectively, as the kernel and cokernel of the canonical homomorphism

$$F \longrightarrow f_* f^*(F) = f_*(F|_{X-A}),$$

and $\underline{H}_A^i(F)$ for $i \geq 2$ is expressed in terms of the derived functors of f_* .

Source note. The French display (28) omits the argument (F) from $\underline{H}_A^1$. The explanatory sentence immediately following the display supplies it; it is restored here and recorded as an emendation.

Corollary 2.12. *Let F be a sheaf of abelian groups on X . If F is flasque, then for every locally closed subset Z of X and every integer $i \neq 0$,*

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$$H_Z^i(X, F) = 0, \quad \underline{H}_Z^i(F) = 0.$$

Conversely, if $H_Z^1(X, F) = 0$ for every closed subset Z of X , then F is flasque.

Suppose that F is flasque. It induces a flasque sheaf on every open subset. To prove that $H_Z^i(X, F) = 0$ for $i > 0$, we may therefore assume that Z is closed; the assertion then follows from the exact sequence (27). For every locally closed Z , “sheafifying”—that is, applying Corollary 2.4—gives $\underline{H}_Z^i(F) = 0$ for $i > 0$. Conversely, suppose that $H_Z^1(X, F) = 0$ for every closed Z . Then (27) shows that $H^0(X, F) \rightarrow H^0(X - Z, F)$ is surjective for every such Z , which means precisely that F is flasque.

Editorial note. At both corresponding occurrences in the French proof, the original edition cited Corollary 2.9; the corrected branch specifies equation (27).

Combining Theorems 2.6 and 2.8, we deduce:

Proposition 2.13. *Let F be a sheaf of abelian groups on X , let Z be a closed subset of X , put $U = X - Z$, and let N be an integer. The following conditions are equivalent:*

- (i) $\underline{H}_Z^i(F) = 0$ for $i \leq N$.
- (ii) For every open subset V of X , the canonical homomorphism

$$H^i(V, F) \longrightarrow H^i(V \cap U, F)$$

is

- a) bijective for $i < N$;
- b) injective for $i = N$.

If $N > 0$, it is enough in (ii) to require a).

To prove (i) $\Rightarrow$ (ii), the local nature of the sheaves $\underline{H}_Z^i(F)$ reduces us to proving the following.

Corollary 2.14. *If condition (i) of Proposition 2.13 holds, then*

$$H^i(X, F) \longrightarrow H^i(U, F)$$

is bijective for $i < N$ and injective for $i = N$.

Indeed, by the exact sequence (27), this is equivalent to $H_Z^i(X, F) = 0$ for $i \leq N$, and this relation is an immediate consequence of the spectral sequence (23 bis).

Editorial note. At both corresponding occurrences in the French text, the original edition cited Theorem 2.6; the corrected branch specifies equation (23 bis).

Conversely, condition (ii) of Proposition 2.13 says that, for every open subset V of X ,

$$H_{Z \cap V}^i(V, F|_V) = 0 \quad \text{for } i \leq N,$$

which implies (i) by Corollary 2.4. Moreover, if $N > 0$, condition b) is superfluous. Indeed, if $N = 1$, condition a) together with (28) gives $H_Z^i(F) = 0$ for $i \leq N$. If $N > 1$, condition a) for $i = N - 1 > 0$, together with (29), gives $H_Z^i(F) = 0$ for $i \leq N$.

Taking Corollary 2.11 into account, this also proves condition (i) of Proposition 2.13.

Remark. Let $Y \rightarrow X$ be a closed immersion, and suppose that locally it has the form $\{0\} \times Y \subset R^n \times Y$. Suppose that F is a locally constant sheaf on X . Then

$$H_Y^i(F) \simeq \begin{cases} 0, & i \neq n, \\ F \otimes \underline{T}_{Y,X}, & i = n, \end{cases} \quad \underline{T}_{Y,X} \simeq H_Y^n(\mathbf{Z}_X), \quad (30)$$

where $\underline{T}_{Y,X}$ is the extension to X of a sheaf on Y locally isomorphic to $\mathbf{Z}_Y$; it is called the *normal orientation sheaf* of Y in X .

Source note. The corrected French prose prints lowercase x where the ambient space must be X , and its sentence defining $\underline{T}_{Y,X}$ is grammatically incomplete. The reading above preserves the display and states its evident mathematical syntax explicitly.

Using the spectral sequence (23 bis), we obtain in this case

$$H_Y^i(X, F) \simeq H^{i-n}(Y, F \otimes \underline{T}_{Y,X}), \quad (31)$$

and recover the *Gysin homomorphism*

$$H^j(Y, F \otimes \underline{T}_{Y,X}) \longrightarrow H^{j+n}(X, F). \quad (32)$$

References

- [Godement] R. Godement, *Topologie algébrique et théorie des faisceaux*, Act. Scient. et Ind., vol. 1252, Hermann, Paris.
- [Tôhoku] A. Grothendieck, “Sur quelques points d’algèbre homologique,” *Tôhoku Math. J.* **9** (1957), 119–221.